

SentimentAI — Project Status Report

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1. Executive Summary

SentimentAI has grown from a single-feature sentiment detection prototype into a **full-stack, multi-channel customer experience platform**. What began as a real-time chatbot that scores emotion in customer messages has expanded into a complete business operations suite covering support, sales, analytics, forecasting, and external communication — all driven by one shared AI engine.

The project is currently built across a **Python FastAPI backend** and a **React (Vite) frontend**, using **Groq's LLaMA 3.3 70B** for all language understanding and generation, and **SQLite** as the data layer.

2. Feature Inventory

2.1 Core Sentiment Intelligence

Feature	Status	Description
Real-time sentiment scoring	 Built	Every message scored 0.0–1.0 with an emotion label
Automatic action routing	 Built	Classifies each message as ESCALATE / NORMAL / UPSELL
Context-aware AI replies	 Built	Bot generates a fresh, tone-matched reply per action type
Persistent conversation memory	 Built	Remembers returning customers by name and history across sessions
Escalation alert system	 Built	Full-screen dramatic alert popup fires on high-frustration messages

2.2 Communication Channels

Feature	Status	Description
Live Chat (WebSocket)	✓ Built	Real-time chat with instant sentiment feedback
Voice Support	✓ Built	Speech-to-text via Groq Whisper, spoken AI replies via Web Speech API
Email Integration	✓ Built	IMAP/SMTP inbox monitoring, AI-drafted replies, auto-process all
Caspian SDK — Email channel	✓ Connected	Live email address (agt-f24c22906bb6a2d589648aa7-a743ed@agents.trycaspianai.com) wired to the same AI pipeline
Caspian SDK — Telegram channel	⚠ In progress	Bot created via BotFather, token generation/connection currently being debugged

2.3 Support Operations

Feature	Status	Description
Ticketing system	✓ Built	Auto-created tickets with sentiment-based priority (High/Medium/Low)
Agent Console	✓ Built	Live handoff queue, agent directory, active conversation view
Human handoff	✓ Built	Auto-routes ESCALATE conversations to the best available agent with full context
Knowledge Base	⚠ Rebuilding	FAQ storage and search exist; integration into the main reply pipeline is currently being debugged so the bot actually uses KB answers before generating fresh ones

2.4 Business Intelligence

Feature	Status	Description
Analytics Dashboard	✓ Built	Sentiment trend, emotion breakdown, action distribution, hourly activity, AI-generated top issues
CRM	✓ Built	Auto-created customer profiles (from chat/email), status tracking, notes, tags

Feature	Status	Description
Customer Profiles	✓ Built	Full activity timeline, AI risk score, AI-generated relationship summary
Deal Pipeline	✓ Built	Kanban-style pipeline (Lead → Qualified → Proposal → Negotiation → Won/Lost)
Predictive Analytics	⚠ Partially working	Churn risk, upsell opportunities, ticket/revenue forecasting built; currently returning a load error being debugged

2.5 Data Portability

Feature	Status	Description
Excel Import	✓ Built	Upload customers/tickets/conversations/deals via Excel or CSV
Excel Export	✓ Built	Export customers, conversations, tickets, and full analytics reports
Templates	✓ Built	Downloadable blank templates for standardized data entry

2.6 Language & Accessibility

Feature	Status	Description
Bilingual UI (EN/HI)	✓ Built	Full interface toggle between English and Hindi
Regional language AI replies	✓ Built	AI detects and replies in Hindi, Hinglish, and other Indian regional languages, matching script and tone

2.7 Platform & Reliability

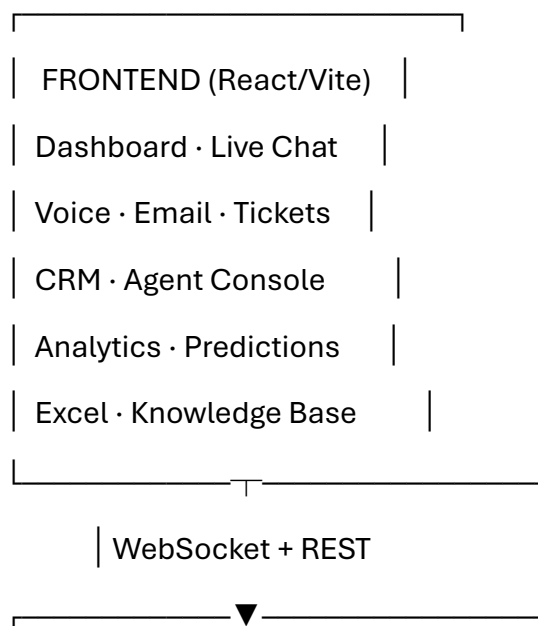
Feature	Status	Description
Live WebSocket connection status	✓ Built	Visible "Connected / Offline" indicator in the header
Global Error Boundary	✓ Built	Crashes now show a readable error and reload option instead of a blank white screen

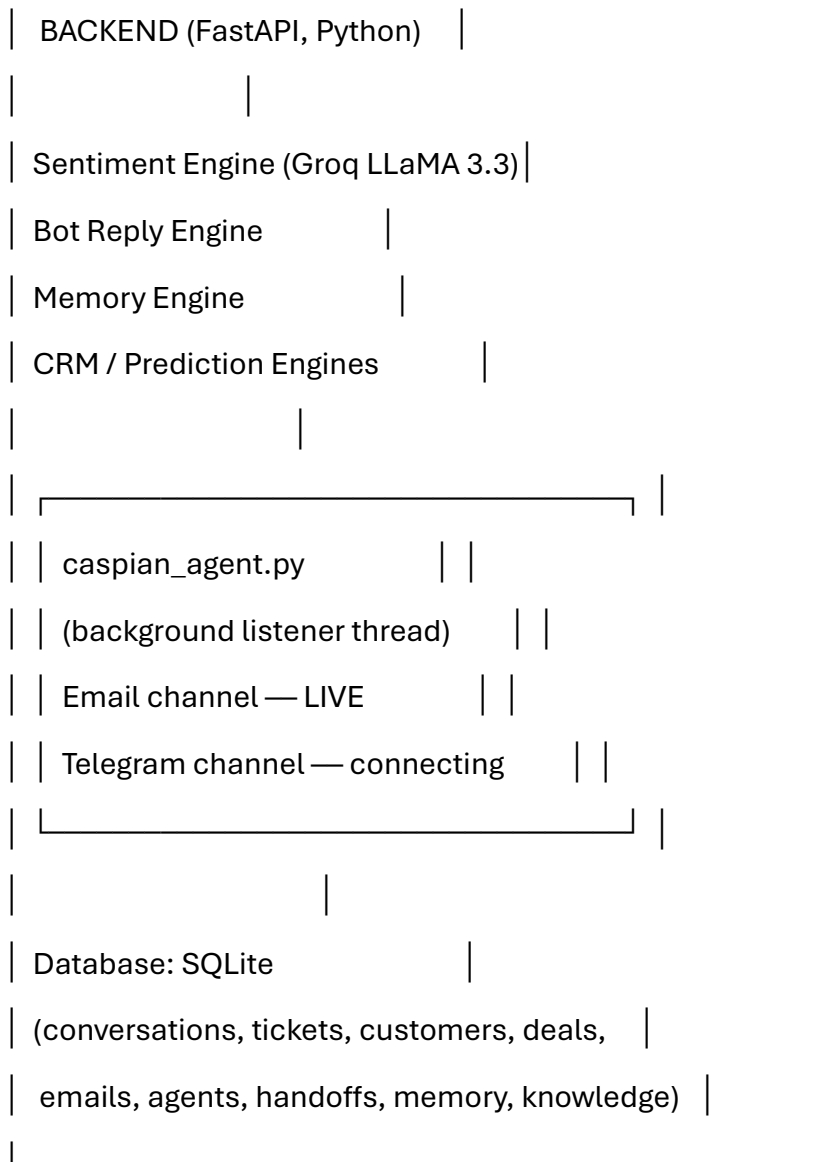
Feature	Status	Description
Dark, real-time dashboard UI	✓ Built	Consistent design system across all modules

3. Currently Known Issues (Active Debugging)

- Knowledge Base not influencing bot replies** — the search function exists, but is not confirmed to be wired into the main `generate_reply()` pipeline used by Live Chat, Voice, and Email. Under active diagnosis with step-by-step logging to isolate whether the issue is missing data, a broken match function, or a pipeline wiring gap.
 - Predictions tab intermittently failing** — the `/predict/dashboard` endpoint throws an error under sparse data conditions (few customers/conversations); needs defensive default handling.
 - Backend startup crash (resolved)** — a mismatched import (`save_conversation_message` vs. the actual function name in `database.py`) was previously causing the entire backend to fail to start. This has been identified and corrected.
 - Telegram channel via Caspian SDK** — bot has been created through BotFather; the connection step (`caspian connect telegram`) is currently being completed.
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4. Architecture Overview





5. Technology Stack

Layer	Technology
AI / LLM	Groq API — LLaMA 3.3 70B Versatile
Speech-to-Text	Groq Whisper (large-v3)
Text-to-Speech	Web Speech API
External Channels	Caspian SDK (email connected; Telegram in progress)
Backend	Python, FastAPI, WebSockets
Database	SQLite

Layer	Technology
Frontend	React, Vite
Data Visualization	Recharts
Email	Python smtplib / imaplib
Spreadsheets	openpyxl, pandas

6. Progress Summary by Development Phase

Phase	Scope	Status
Phase 1	Sentiment engine, WebSocket, live dashboard, escalation alerts	✓ Complete
Phase 2	Bot auto-reply, SQLite persistence, conversation history	✓ Complete
Phase 3	Bilingual UI, dramatic alert system	✓ Complete
Phase 4	Ticketing, Live Chat, Knowledge Base (initial)	✓ Complete (KB re-integration in progress)
Phase 5	Analytics Dashboard	✓ Complete
Phase 6	Voice support (STT + TTS)	✓ Complete
Phase 7	Email integration (IMAP/SMTP)	✓ Complete
Phase 8	CRM, deal pipeline, customer profiles	✓ Complete
Phase 9	Agent Console, human handoff	✓ Complete
Phase 10	Auto-CRM creation from chat	✓ Complete
Phase 11	Predictive Analytics	⚠ Built, stabilizing
Phase 12	Excel import/export	✓ Complete

Phase	Scope	Status
Phase 13	Regional Indian language support	✅ Complete
Phase 14	Persistent memory + voice/chat unification on Dashboard	✅ Complete
Phase 15	Caspian SDK integration (multi-channel external messaging)	⚠️ In progress (Email live, Telegram connecting)

7. Next Steps

1. Complete Knowledge Base wiring into the main reply pipeline with verified proof (logs showing KB match → reply used).
 2. Stabilize the Predictions endpoint against sparse data.
 3. Complete Telegram channel connection via caspian connect telegram.
 4. End-to-end test of both live Caspian channels (real email + real Telegram message in, AI reply out, ticket auto-created on escalation).
 5. Final documentation pass (README, feature document, demo script) once the above are stable.
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Summary: The core AI engine, dashboard, and majority of business modules (CRM, Tickets, Analytics, Agent Console, Voice, Email, Excel, Multi-language) are fully functional. The remaining work is concentrated in two areas — Knowledge Base pipeline wiring and Predictions stability — plus completing the Telegram leg of the Caspian multi-channel requirement.